Answers to NJC H2 Biology Preliminary Paper 3

Section A

1. A reporter gene is a gene whose activity is relatively easy to detect. Luciferase is an enzyme naturally found in fireflies. When its substrate luciferin is present, it catalyses a chemical reaction which emits light, as shown by the equation below. Therefore, luciferase gene can be used as a reporter gene in recombinant technology.

luciferin + ATP + oxygen → oxyluciferin + AMP + light

Fig. 1.1 shows a recombinant plasmid construct named "pMCS-Red Firefly Luc" used to study the regulation of *lac* operon in bacteria.

(a) State the factors which you will need to consider when choosing the right restriction enzyme to construct the pMCS-Red Firefly Luc. [2]

- The restriction site of the enzyme must be present in both the MCS of the plasmid and the sequence flanking the Red Firefly Luc insert, so that complementary sticky ends will be produced for ligation.
- The restriction site must only be found once in the whole plasmid / must be a unique cloning site.

(b) Both genomic and cDNA libraries of firefly are available. State where you will take the Luciferase gene from for this experiment. Justify your decision. [2]

- cDNA libraries
- Firefly is a eukaryote whose genes contain introns which need to be removed through splicing during post-transcriptional modifications of pre-mRNA. Bacteria are prokaryotes which do not perform post-transcriptional modifications. Hence, cDNA of luciferase gene which only contain exons is needed for bacteria to produce functional luciferase.

(c) Plasmids are introduced into bacteria via electroporation. [2]

(i) Outline how you are to select bacterial cells transformed with recombinant plasmid containing the Red Firefly Luc insert.

- Grow the bacteria in an agar medium containing ampicillin / puromycin, those surviving colonies are the ones transformed with plasmids.
- Add IPTG/ lactose (served as inducer to switch on the operon) and luciferin (substrate of luciferase) to the medium. Those colonies which give out luminescence are cells transformed with recombinant plasmids containing the luciferase gene.

A few experiments are conducted to study the regulation of *lac* operon via expression of luciferase gene in the bacterial colonies.

Complete the table below by putting in '+' (low light intensity), '++' (high light intensity) and '-' '(no light emitted) in the last column.

Experiment	Glucose	Lactose	Ampicilin	Luciferin	Brightness of colonies
1	✓	×	✓	✓	-
2	✓	✓	✓	✓	+
3	×	✓	✓	✓	++
4	×	✓	✓	×	-
5	×	×	✓	✓	-

[2]

(ii) Explain your results in experiment 1 and 3. [4]

- In the absence of lactose, constitutively expressed lac I gene produces active repressor protein which binds to the operator, preventing RNA polymerase from transcribing the luciferase gene.
- No light is emitted when there is no luciferase catalyzing the reaction in experiment 1.
- When lactose is present, allolactose acts as an inducer which binds and changes the conformation of repressor protein. Thus the repressor protein can no longer bind the operator, allowing RNA polymerase to transcribe the luciferase gene.
- In the absence of glucose, there is high cAMP concentration in bacterial cell, leading to formation of active cAMP-CAP complex binding to CAP binding site, increasing the affinity of RNA polymerase for promoter and thus increasing transcription rate of the luciferase gene.
- High amount of luciferase causes more light to be emitted in experiment 3. (combined marking point with 2nd bullet point

(d) Instead of Red Firefly Luc, human insulin gene can also be cloned into pMCS plasmid shown in Fig.1.1. Suggest a suitable method to select bacterial cells transformed with recombinant plasmid containing the insulin gene insert. [2]

- Grow bacteria on medium containing ampicilin / puromycin to select those transformed with pMCS plasmid.
- To select colonies transformed with recombinant plasmid, add radioactively/fluorescently labeled single stranded nucleic acid probe complementary to the insulin cDNA sequence and perform autoradiography / view under UV light.

[Total: 14]

2. Xeroderma pigmentosum-variant (XP-V) is a autosomal recessive disorder caused by defective DNA repair. Individuals with this disorder have sun sensitivity and increased skin cancer risk. One of gene identified is the POLH gene, which encodes for DNA polymerase (pol).

The molecular basis of XP-V worldwide was analyzed from a family in North America. Insertion of a single guanine at position 1078 of exon 10 creates a new *Bsl1* site as shown in Fig. 2.1 and is linked to an RFLP.

(a) State and explain the possible consequence of such gene mutation. [2]

- Leads to frame-shift mutation as the bases downstream are re-grouped into different codons
- Extensive missense or nonsense mutations can be resulted → Causing different sequence of amino acid coded or shorter amino acid sequence
- Altered three dimensional structure of DNA polymerase / truncated protein leading to a loss of biological function

(b) Suggest two factors that are required for a RFLP marker to be used in disease detection. [2]

- The marker must be polymorphic, meaning that alternative forms must exist among individuals for detection among different members in the family.
- The marker must be closely linked to the disease locus such that crossing over between the marker and the gene is unlikely to occur during gamete formation. The greater the distance between the RFLP and gene locus, the lower the probability of an accurate diagnosis.

(c) PCR amplification of the entire coding region of POLH was first performed.

With reference to Fig. 2.2, comment on the purpose of:

(i) Step A [1]

- Annealing of primers / oligonucleotides to single DNA strands at 3’ end via complementary base pairing

(ii) Step B [1]

- DNA strand elongate from the 3’ end of the DNA primer using <u>Taq DNA polymerase</u> where <u>extension of DNA strand</u> occurs from <u>5’ to 3’ direction</u>

Digestion of POLH gene from individuals within the family was then carried out using *Bsl1*. Fig. 2.3 shows the incomplete pedigree chart and the results obtained from gel electrophoresis.

(d) Using the information provided in the question and with reference to Fig. 2.3, state and explain the genotype of individual III-6. [2]

- Genotype for III-6: Hh
- Have the presence of one band that corresponds to the wild-type allele which is 353bp, while the other two bands correspond to the mutant allele which corresponds to 105bp and 248bp

(e) Outline the experimental procedure, after the action of Bsl1, of obtaining a DNA fingerprint. [4]

- DNA restriction fragments are size-fractionated by gel electrophoresis;
- Bands obtained are transferred / blotted from the gel onto a nitrocellulose membrane
- The DNA is denatured into single strands by the alkaline buffer
- Membrane is then incubated with radioactively / chemically labeled single-stranded DNA probes which hybridize to complementary target sequences on the membrane
- The membrane is washed thoroughly to remove unbound probe;
- This is followed by detection of the bound probe by autoradiography / chemical means to produce a specific pattern of bands.

(f) Apart from disease detection, explain how RFLP analysis facilitates genomic mapping. [3]

- [RFLP as genetic marker]: RFLP markers spaced throughout the chromosomes are used as genetic markers (such as STRs, and other polymorphic sites)
- [RFLP in the construction of linkage map]: Frequency with which two RFLP markers or an RFLP marker and a certain allele for a gene are inherited together is a measure of the closeness of the 2 gene loci on a chromosome.
- The higher the frequency of 2 RFLP being inherited together, the closer the 2 loci are on the chromosome / Based on cross-over values / recombination frequency / frequency of crossing-over between the RFLP marker and the gene loci (or between 2 RFLP markers)
- [RFLP in determining the order of markers]: The presence of RFLP on a chromosome can serve as marker to order the sequence of genes and their relative distances on a chromosome.

[Total: 15]

3. **(a)** Immature anther of a hibiscus flower bud was removed and the following main procedures were introduced in the making of a complete plantlet.

(i) Explain why it is important to sterilise the anther tissues. [1]

- This is to prevent contamination/ growth of microorganisms such as fungi and bacteria, which will outgrow the callus / prevent callus formation/compete for nutrients

(ii) Describe briefly the importance of adding sucrose in the sterile medium. [1]

- Sucrose acts as a carbon and energy source
- as plant cells in the culture medium lack autotrophic ability/ are unable to photosynthesize. **or**
- to enhance proliferation of cells and regeneration of shoots.

(iii) Explain the purpose of adding the chemical colchicine in the final stages of the procedure. [2]

- To double the chromosome number, to restore the haploid cells back to the diploid condition.
- By preventing mitotic spindle formation
- Hence sister chromatids cannot be separated during anaphase of mitosis

(b) Explain the advantages of tissue culture over more traditional methods of cloning plants, such as taking cuttings or graftings. [2]

- The rapid and mass multiplication of plants is made possible because a single plant can be cloned into thousands or millions of genetically identical copies by subdividing calluses as they grow.
- Large numbers of disease-free plants may be produced if explants from meristematic regions are used. This is called meristem culture. The meristematic regions of a plant are the shoot tips and root tips. The meristem is free from bacteria and viruses because the latter either do not easily invade or do not rapidly multiply in the meristematic tissue.
- Plant tissue cultures take up relatively little space compared to growing plants in greenhouses or on farms.
- Micropropagation overcomes the barrier posed by seasons in conventional propagation methods. Therefore, plants can be produced out-of-season to meet consumer demand.
- Micropropagated plants can be airfreighted easily and cheaply, because cultures are not bulky.
- Micropropagation makes it possible to standardize growth conditions, and obtain many batches of identical plants. This ensures product uniformity for customers.
- Many genetically engineered plants can be multiplied and mass produced relatively easily.

(c) With relevant example, explain how genetic engineering has improved the quality of crops. [3]

- Flavr Savr Tomato – antisense technology
- Gene inserted which produces mRNA that is complementary to the mRNA of polygalacturose
- Double stranded mRNA formed and ribsome binding prevented, translation cannot occur
- Delayed ripening and hence taste, texture improves

- Or Golden rice as example.

(d) Researchers are experimenting with the injection of human brain cells into the brain tissues of mice. Suggest the ethical considerations that may arise. [2]

- Whether humans ought to be creating living organisms not envisaged by nature
- It is important that animal chimeras not develop uniquely human characteristics such that some degree of humanity or human dignity has been conferred on them. This can lead to a sense of moral confusion and could create obligations toward these new entities that would require rethinking our categories of animals and humans.
- Violating the genetic integrity of species, that has been a result of many generations of evolution.
- Others find the alteration of natural physical characteristics the source of their unease or repugnance.
- If the goal of the unnatural process is immediate and life-saving therapy it might be more morally acceptable than research of some future indeterminate benefit.

[Total: 11]

4. Planning question

Write your answers to this question on the separate answer paper provided.

A type of genetically modified sugar cane contains higher amount of sucrose per unit mass. You are required to measure the sucrose concentration of freshly squeezed sugar cane juice from this new variant.

Your planning must be based on the assumption that you have also been provided with the following materials which you must use:

- A beaker of sugar cane juice labelled as **S**
- a 10% sucrose stock solution
- distilled water
- Dilute hydrochloric acid
- Sodium hydrogen carbonate solution
- Thermostatically controlled water bath
- Benedict's solution
- Colorimeter

In addition, you may include in your plan the usage of any equipment that can be found in a standard school laboratory.

Your plan should have a clear and helpful structure to include:

- An explanation of the theory to support your practical procedure.
- A description of the method used, including the scientific reasoning behind the method
- An explanation of the dependent, independent variables and controlled variables
- How you will record your results and ensure that they are as accurate & reliable as possible.
- Proposed layout of results tables and graph with clear headings and labels
- Recommended safety measure.
- The correct use of technical and scientific terms.

[Total: 12]

Suggested answers for Planning Question

Explanation of theoretical background (2 marks)

- Sucrose is a non-reducing disaccharide.
  When the glycosidic bond in sucrose is hydrolysed by dil. HCl under heat, glucose and fructose is produced. Both glucose and fructose are reducing sugar.

- When Benedict's solution containing Cu2+ is added, reducing sugar causes Cu2+ to be reduced to Cu2O, which is a red precipitate.

Method and scientific reasoning (can be mentioned in the <u>procedure</u> also)

- Dilute hydrochloric acid is used to hydrolyse sucrose into reducing sugars glucose and fructose

- Benedict's solution is then added to test for the amount of reducing sugars / glucose and fructose present.

Hypothesis (can be mentioned in <u>prediction and explanation of results</u> also) (2 marks)

- The higher the concentration of sucrose, the greater the amount of brick-red precipitate produced from the Benedict's test, and the less Cu2+ remaining in the Benedict's solution

- After filtering the ppt out, use colorimeter to measure the colour intensity of the filtrate, and lower absorbance value is expected.

- Sucrose solutions of known concentration are measured for absorbance values in colourimeter

- Compare the absorbance value of the **S** with absorbance values of sucrose solutions of known concentrations by reading off the graph plotted

Variables: (1 mark)

<u>Independent variables:</u> sucrose concentration (%)

<u>Dependent variables:</u> amount of ppt produced in Benedict's test measured as absorbance value in colorimeter (A.U.)

<u>Controlled variables (can be mentioned in procedure also)</u>
- volume of sucrose solution used
- volume of HCl used and duration of hydrolysis reaction
- volume of Benedict's solution added and duration of boiling
- Temperature of hydrolysis reaction and Benedict's Test

Experimental procedure (max 5)

- Using simple dilution (include **<u>dilution table)</u>** to obtain **<u>at least five</u>** sucrose solutions of known concentrations, with **<u>equal intervals</u>** (concentration range: 1% - 10%)

- Use syringe to add 2 cm^3 of each sucrose solution (including solution S) into a separate test tube
- Use syringe to **<u>add 2cm^3 dilute hydrochloric acid</u>** to test tube and mix well

- Place the test tubes into a thermostatically controlled water bath set at 100^oC / **<u>boiling water bath</u>** for three minutes (accept reasonable time)

- Remove the test tubes from water bath. Use syringe to add $5cm^3$ (accept reasonable range) of sodium hydrogen carbonate solution until no effervescence is observed. This is to quench the acid hydrolysis reaction by neutralising the acid

- Use syringe to add **<u>$2cm^3$ Benedict's solution</u>** to each test tube

- Place test tubes into a thermostatic water bath set at **<u>100^oC / boiling water bath</u>**; **<u>for 2 minutes</u>** (accept reasonable time);

- Remove the test tubes and filter the ppt out.

- Place a small sample of each filtrate into the colorimeter to read the absorbance value

- Conduct **<u>3 replicates</u>** for each sucrose solution;

- Repeat the entire experiment twice to ensure reproducibility;

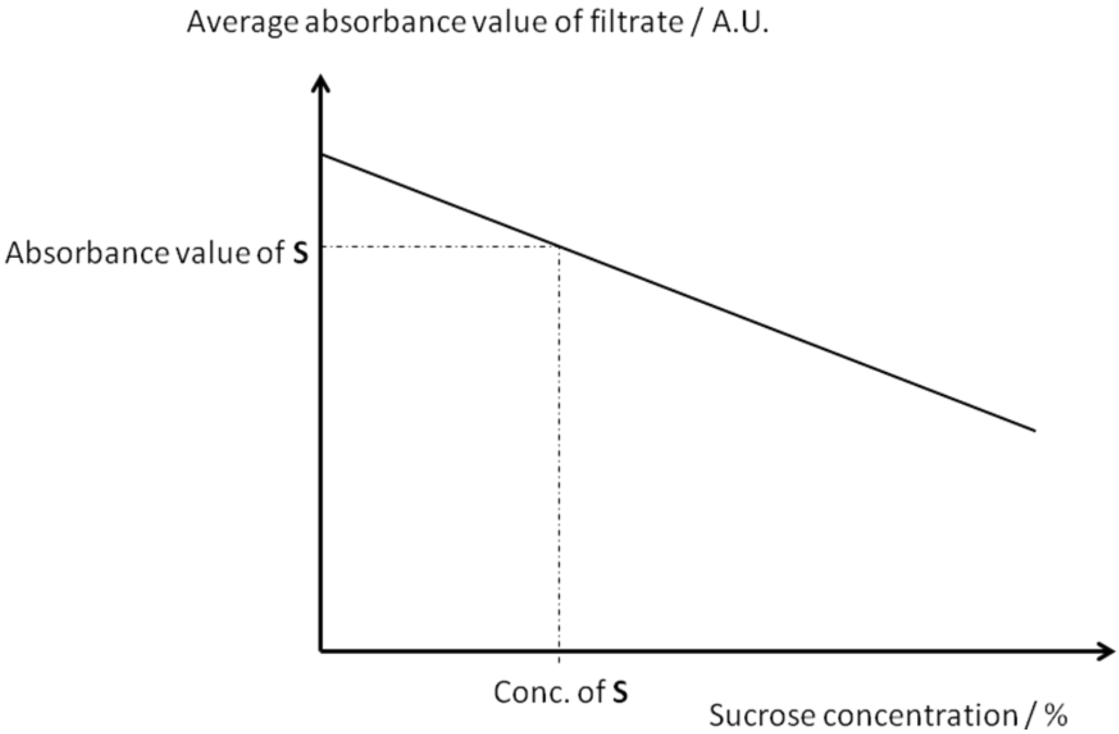
Data recording (2 marks)

- Tabulation of data with correct column headings and units; include average value for absorbance value;

sucrose concentration / %	absorbance value of filtrate / A.U.			
	Exp 1	Exp 2	Exp 3	Average
2				
4				
6				
8				
10				
Solution S				

- Plot a graph of average absorbance value against sucrose concentration;

- Correct labelling of x and y-axis;



Safety precaution *(1 mark)*

- Reagents such as Benedict's solution, dilute hydrochloric acid, etc are corrosive to skin and eyes. Wear gloves and goggles. Rinse under running water when in contact occurs.

- Boiling water bath can scald. Handle hot glassware such as test tubes with test-tube holders or oven gloves/rags.

Free Response Question

Write your answers to this question on the separate answer paper provided.

Your answer:

- Should be illustrated by large, clearly labelled diagrams, where appropriate;
- Must be in continuous prose, where appropriate;
- Must be set out in sections (a), (b), etc., as indicated in the question.

5 **(a)** Explain, with examples, the treatment of genetic diseases with reference to viral and non-viral gene delivery system. [8]

(b) Discuss the social and ethical concerns that have arisen from gene therapy in the treatment of genetic diseases. [6]

(c) Discuss on the different types of stem cells. [6]

(a) Explain, with examples, the treatment of genetic diseases with reference to viral and non-viral gene delivery system. [8]

1. Gene therapy is a technique for introducing normal allele for the purpose of generating functional (correct) protein product to restore the target cell to normal state (correct function of cell / correct phenotype)

2. Example of viral gene delivery method involves SCID treatment, using ex vivo method

3. Functional ADA allele is cloned into retroviral vector. The viral vector has to be genetically engineered so that they neither produce disease nor replicate in the host cell.

4. Patient's lymphocytes will then be extracted from the blood and be transfected with ADA recombinants
5. Following infection, the recombinants cells can be cultured in a suitable medium and suitable cells can then be expanded in culture in vitro before they were re-injected into the patient's blood stream.

(Max: 4)

6. Example of viral gene delivery method involves CF treatment, using in vivo method

7. Functional CFTR allele is cloned into adenoviral vector. The viral vector has to be genetically engineered so that they neither produce disease nor replicate in the host cell.

8. Recombinant adenovirus must be delivered effectively to introduce a functional CFTR allele into the nucleus of the epithelial cells lining the bronchial tree within the lungs

9. Directly either through a bronchoscope or through the nasal cavity. → the gene is inserted into cells within body

10. Example of using non-viral gene delivery method involves cystic fibrosis treatment, using the in vivo method

11. In vivo cationic liposomes are used and consists of lipid : DNA complexes that were developed for the transfection of epithelial cells lining the airway

12. DNA (functional CFTR allele) interacts with the cationic lipids on the inner layer via electrostatic interactions. The cationic lipids on the outer layer serve to play a role in enhancing liposome interaction with the target cells

13. The functional CFTR allele enters the target cells (i.e. lung epithelial cells) through the lipids fusing with the plasma membrane, and the DNA eventually entering the cell nucleus. → direct introduction of DNA into target cells

14. Liposomes containing the functional CFTR allele are administered by the inhalation of aerosols by the patient.

(Max: 4)

(b) Discuss the social and ethical concerns that have arisen from gene therapy in the treatment of genetic diseases. [6]

<u>Social considerations</u>

1. Gene therapy is currently very expensive and there is concern about gene therapy becoming accessible only to the wealthy

2. Possible enhancements in creating an advantage for those who can afford the treatment

3. Large numbers of gene therapy trials are currently directed to diseases with an enormous social impact, such as cancer, SCID, CF etc.

4. Gene therapy might provide new forms of treatments directed to currently unmet clinical needs with the potential impact to improve patient's quality of life

5. Concern over widespread use of gene therapy making society less accepting of people who have mild disorders

6. Potential for non-therapeutic enhancement possibilities / eugenic social policies

7. Gene therapy might provide new forms of treatments directed to currently unmet clinical needs / alternative treatments for patients where conventional treatments have failed.

<u>Ethical considerations</u>

8. D<u>ifficulty in following-up with patients in long-term clinical research</u>. Gene therapy patients have to be under surveillance to monitor for long term effects of the therapy on future generations.

9. Concerns over whether patients and their family members fully understand the risks associated with gene therapy trials

10. And whether they make informed decisions before participating in a trial

11. Ref. to Jesse Gelsinger (1st death) gene therapy trial → he had a relatively mild form of a disease that was being successfully treated with medication and perhaps should not have been a gene therapy trial participant

12. Issues pertaining to safety and efficacy of gene therapy as such therapy have not yet had the success anticipated by many scientists as a few unexpected complications and side effects have been identified. Therefore, the novelty of risks related to gene transfer means there will be lots of uncertainties as compared to conventional therapeutics.

13. Raises questions over who should participate in a human trial – the terminally ill with no treatment options or anyone who understands the potential risk but is a willing participant

14. Sponsors of human clinical trials must exercise responsibility in balancing experimental aims and ensuring that participants are not exposed to known risks / experimental products are as safe as possible.

15. Issues of justice and allocation where the high cost of gene therapy for the actual treatment and technical expertise may leads to inequality between individuals with genetic enhancements and those without.

16. In the event of disastrous outcomes brought about by gene therapy, the challenge lies in identifying who should be held accountable

17. Germ-line gene therapy will violate the rights of subsequent generations to inherit a genetic endowment that has not been intentionally modified and permission must be sought from the individuals in these generations as to whether are they willing to undergo gene therapy.

18. Concerns about the protection of privacy and confidentiality of medical information of patients involved in clinical trials

19. Which may have implications on (medical) insurance coverage and/or employability

20. There can be the potential harmful abuse of the technology where individuals are using the technique for enhancement rather than for therapeutic purpose. → Tampering with human genes / nature and bias handling of trial results

(c) Discuss on the different types of stem cells. [6]

1. Zygotic stem cells [Example : Fertilised zygote]

- These are the cells in morula, which are formed during the first few divisions after fertilisation/fusion of an egg and sperm.
- Zygotic stem cells, which include the zygote and early embryo cells are totipotent.
- They have the ability to differentiate into any cell type to form whole organisms and hence are also pluripotent and multipotent.

Functions

2. Zygotic stem cell has the potential to differentiate into all the cells that form the complete organism plus the extra-embryonic tissues which include the placenta and amnion.

3. These cells give rise to form all cells in the body of the organism therefore they have limited ability to self-renew.

4. Embryonic stem cells

- Embryonic stem cells are pluripotent. They have the ability to differentiate into almost any cell type to form any organ or type of cell and so are not totipotent but are multi-potent.

- They are obtained from the Inner Cell Mass on the fourth day of development where the embryo forms into two layers, an outer layer (extra-embryonic membranes) and an inner cell mass (ICM).

Functions

5. Inner cell mass (ICM) forms the three primary germ layers: ectoderm, endoderm and mesoderm, which later form all the tissues of the human body.

6. Give rise to various organs in organism / multiple specialized cell types that make up the heart, lung, skin and other tissues in the developing foetus.

7. Adult stem cells [E.g. Blood stem cells, Stromal Stem cells, Gut Epithelium stem cells]

- Blood stem cells are multipotent. / found in each tissues or organs which have the ability to differentiate into a limited range of cell type and so are not pluripotent or totipotent.

- Found in bone marrow or blood/skin etc

Functions

8. Help to replenish worn-out or damaged cells / responsible for constant renewal of blood

9. Give rise to all the blood cell types

10. Properties pertaining to stem cells

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